

- Chapter 3: SYSTEM ANALYSIS
 - 3.1 Overview
 - 3.2 Existing System Analysis
 - 3.2.1 The Static Paradigm (Traditional LMS)
 - 3.2.2 The Stochastic Paradigm (Pure LLM Tutors)
 - 3.3 Problem Statement
 - 3.4 Objectives of the Proposed System
 - 3.5 Proposed System Analysis: The Hybrid Solution
 - 3.6 Feasibility Study
 - 3.6.1 Economic Feasibility
 - 3.6.2 Technical Feasibility
 - 3.6.3 Operational Feasibility
 - 3.7 Summary

Chapter 3: SYSTEM ANALYSIS

3.1 Overview

System analysis is a systematic process of examining an existing situation, identifying its deficiencies, and defining the requirements for a new, improved system. For CareerOrbit, the analysis phase is critical because it justifies the transition from traditional, linear LMS models to a complex, multi-agent hybrid architecture. This chapter evaluates the current market state, formalizes the research problems, and conducts a multi-dimensional feasibility study.

3.2 Existing System Analysis

Current digital learning environments generally fall into two categories, both of which demonstrate significant systemic failures when applied to vocational training for marginalized demographics.

3.2.1 The Static Paradigm (Traditional LMS)

Platforms such as Coursera, Udemy, and edX rely on pre-recorded video sequences and static quizzes.

- **Inflexibility:** The instructional pace is fixed. A learner who struggles with a foundational concept (e.g., Ohms Law) is pushed forward to complex tasks (e.g., Circuit Troubleshooting), leading to a collapse in confidence.
- **Data Blindness:** These systems track "completion" rather than "mastery." A user can pass a quiz by guessing, but the system treats it as genuine knowledge.
- **Motivation Collapse:** Without real-time intervention, the learner is left alone in the "Valley of Despair," resulting in the industry-wide 90% dropout rate.

3.2.2 The Stochastic Paradigm (Pure LLM Tutors)

Recent attempts to use AI (like early ChatGPT tutors) rely solely on large-scale probabilistic text generation.

- **Cognitive Amnesia:** Pure LLMs do not have a persistent, mathematically-bounded model of the user. They "forget" what the user has mastered from one session to the next.
- **AI Drift:** Without a deterministic controller, the AI often wanders off-topic or provides instruction that is either dangerously complex or patronizingly simple.
- **Pedagogical Violations:** They frequently violate the ZPD by providing direct answers instead of providing the necessary scaffolding to lead the learner to the answer.

3.3 Problem Statement

The central research problem addressed by CareerOrbit is: **"How can we provide instructionally rich, generative tutoring that remains mathematically grounded and strictly aligned with a learner's Zone of Proximal Development (ZPD)?"**

Secondary problems include:

1. **The Guess/Slip Ambiguity:** How to distinguish between a lucky guess and genuine mastery in a digital interface.

2. **The Context Window Limitation:** How to maintain a long-term "Cognitive Memory" of a user across thousands of learning interactions without exceeding LLM context limits.
 3. **The Language Accessibility Gap:** How to deliver high-depth technical training in regional languages while ensuring the AI tutor remains within the bounds of National Occupational Standards (NOS).
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3.4 Objectives of the Proposed System

The proposed system, CareerOrbit, aims to achieve the following specific technical and pedagogical objectives:

1. **Implementation of a Hybrid BKT-IRT Logic:** To create a mastery engine that accounts for both the learner's history and the item's difficulty.
 2. **Deterministic Scaffolding Control:** To use the BKT output ($pMastery$) as a hard constraint for LLM-generated instructions.
 3. **Real-time Cognitive Recalibration:** To enable the system to "rewrite" learning paths instantly if a learner demonstrates unlearning or regression.
 4. **Cinematic Mastery Visualization:** To provide the user with a "BKT-First" dashboard that visualizes their digital twin's knowledge growth.
 5. **Research-Grade Telemetry Integration:** To log all interactions for the calculation of Hake Gain and SDG 8 impact metrics.
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3.5 Proposed System Analysis: The Hybrid Solution

The proposed system operates on a **Three-Layer Hybrid Architecture**:

- **Layer 1 (The Anchor):** The Bayesian Knowledge Tracing (BKT) engine provides the ground truth. It is a deterministic mathematical model that is immune to hallucination.
- **Layer 2 (The Voice):** The Large Language Model (LLM) provides the pedagogical dialogue. It is grounded by the BKT state through "Contextual Injection."

- **Layer 3 (The Feedback Loop):** The User Interface (Next.js) captures high-fidelity behavioral signals (time-on-task, clicks, quiz results) and feeds them back into Layer 1 to refine the model.
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3.6 Feasibility Study

A rigorous feasibility study was conducted to ensure that CareerOrbit is viable for large-scale deployment.

3.6.1 Economic Feasibility

- **Inference Costs:** While GPT-5.4 reasoning is expensive, CareerOrbit uses a tiered model strategy. Complex roadmap generation uses T1 models, while instant mentor chat uses T2 (gpt-4o-mini), reducing operating costs by 70%.
- **Value Proposition:** The cost of one-on-one human tutoring is $\approx ₹500/hr$. CareerOrbit provides equivalent 2σ gain for $< ₹1/hr$ in API costs, making it highly feasible for vocational councils and NGOs.

3.6.2 Technical Feasibility

- **Stack Scalability:** The use of **Neon Serverless PostgreSQL** and **Vercel Edge Functions** ensures the system can scale from 1 to 100,000 users without manual infrastructure management.
- **AI Maturity:** Current LLMs (GPT-4 class) have reached the "Instruction Following" maturity required to obey complex BKT-derived system prompts.

3.6.3 Operational Feasibility

- **User Literacy:** The UI is designed for users with low digital literacy, focusing on "Mastery Orbs" and voice input (**SpeechInput**) to reduce friction.
 - **Low-Bandwidth Optimization:** The system prioritizes text-based instruction and lightweight JSONB updates, ensuring it remains functional on 3G/4G networks in rural areas.
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3.7 Summary

The system analysis confirms that existing static and stochastic models are insufficient for serious vocational training. CareerOrbit's Hybrid LLM+BKT model is not only technically feasible but also economically and operationally viable, providing a scalable solution to the global skills gap.